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Summary

Systems engineer working in Rust and C on Linux, with hands-on network protocol work: IPP (over HTTP) and DNS-SD discovery in the COSMIC printers backend, and an IPv4 multicast socket option merged into the Zephyr RTOS networking stack. Google Summer of Code 2026 contributor with The Linux Foundation (OpenPrinting): landed 7 PRs upstream in cups-rs and libcups.

Experience

Google Summer of Code Contributor May 2026 – Oct 2026
The Linux Foundation (OpenPrinting) *Remote*

- Built the **Printers page for COSMIC Settings** and a standalone app: printers grouped by device, supply levels, print queues, live updates and an Add Printer flow for legacy printers via PAPPL (cosmic-settings #2182, in review).
- Ran the backend as a varlink IPC service in `cosmic-settings-daemon` (cosmic-settings-daemon #188, in review), so the C FFI code never loads in the Settings process.
- Got **7 PRs merged upstream** in OpenPrinting. cups-rs (Rust): ported to libcups3 with CI, IPP management, Printer Application support and DNS-SD FFI bindings (PRs #18–#22). libcups (C): fixed DNS-SD crashes and a leaked client (#164, #165).

Embedded Software Engineer Aug 2025 – Present
EKSON *Egypt*

- Wrote the ESP32-S3 firmware and migrated it from ESP-IDF/FreeRTOS to Zephyr RTOS: PID valve control with IMU feedback, safety stops and a UART protocol.
- Developed the PyQt5 desktop control app, cutting CPU usage and latency by about 40% (measured) using asyncio to wait for ESP32 and server messages concurrently.

Embedded Linux Systems Intern Dec 2024 – Jan 2025
Orange Digital Center (ODC) *Egypt*

- Created custom Linux images with Yocto and Buildroot and wrote a Linux kernel module; worked with device trees, root file systems and hardware flashing. Scored 100% in the final evaluation.

OSS Projects

Zephyr RTOS, Boa JavaScript engine (Rust), Rust Clippy | *Contributor (all merged)* Mar 2025 – Dec 2025

- Zephyr RTOS: added the `IP_MULTICAST_LOOP` socket option for IPv4 in 6 networking modules (#86977).
- Boa: optimized `JsStr::code_points()` for Latin1 strings, avoiding UTF-16 conversion (PR #4553).
- Rust Clippy: fixed `ref_as_ptr` false positives for non-temporary references in `let/const` (PR #16214).

Projects

Kernel Monitor | *C, Linux, ARM, QEMU* GitHub

- Character device driver that walks the process list under RCU and exports per-task IDs, memory, threads, context switches and CPU time to user space; run on QEMU.

Technical Skills

Languages: Rust, C, Modern C++, Python (asyncio), Bash, JavaScript (Node.js)
Systems & Linux: Linux (Ubuntu/Debian), operating systems, FFI and bindings, IPC (varlink, D-Bus; zbus, zlink), async and multithreaded code, memory management, kernel modules and device drivers
Networking: TCP/UDP sockets, IPv4 multicast, HTTP/REST, IPP, DNS-SD/mDNS (Avahi), CUPS
Tooling: Cargo, Clippy, rustfmt, Git/GitHub, GitHub Actions CI, CMake, Make, Yocto, Buildroot, QEMU, debugging, code review

Education

Minia University Minia, Egypt
Bachelor of Science (B.Sc.), Computer and Systems Engineering *2020 – 2025*

- Grade: Very Good with Honors.
- Activities: Codeforces Expert; 4-time ECPC finalist (2021–2024); Treasurer, IEEE Student Branch.